

STATISTICS 801

Exam 2 (100 points)

Name key

Instructions:

- You are allowed 110 minutes for this exam. Please pace yourself accordingly.
- You may use a calculator and your formula sheet on this exam. However, you are not allowed to use your cell phone AT ALL! It is a "closed book" exam.
- The t-table and the Studentized range q-table are attached to your exam so you DON'T need to use your own tables.
- In order to receive full credit, you must show your work and carefully justify your answers. The correct answer without any work will receive little or no credit.
- If you need more room, use the backs of the pages AND INDICATE that you have done so.
- Please write neatly.
- GOOD LUCK!!

1. An experiment is conducted to compare a revolutionary new method of teaching reading to "slow learners" to the current standard method. Of a random sample of 20 slow learners, 8 are taught by the new method and 12 are taught by the standard method. Both groups were taught by qualified instructors and under similar conditions for a 6-month period. The scores obtained by the participants in a reading test given at the end of this period, and some relevant statistics computed for the two samples, respectively, are given below:

New Method				Standard Method			
80	80	79	81	79	62	70	68
76	66	79	76	73	76	86	73
				72	68	75	66
$n = 8$				$n = 12$			
$\bar{y}_1 = 77.125$				$\bar{y}_2 = 72.333$			
$s_1^2 = 23.55$				$s_2^2 = 40.24$			

Let μ_1 and μ_2 represent mean reading scores of slow learners taught using the new method and the standard method, respectively. Assume that a larger score implies an improvement.

- (a) (10) Is there a reason to believe that the assumption of equal population variances is justified? Explain. Perform a statistical test to verify this assumption, stating the hypotheses and the rejection region clearly. Use $\alpha = .05$.

For making your decision use the appropriate F -value(s). Circle the one(s) you are using!

$F_{0.025,7,11} = 3.76$ or $F_{0.025,8,12} = 3.51$ or $F_{0.025,1,8} = 7.57$ or
 $F_{0.025,11,7} = 4.7$ or $F_{0.025,12,8} = 4.2$ or $F_{0.025,8,1} = 956.7$.

Yes, since s_1^2 and s_2^2 differ by a factor of less than 2, there is no reason to believe that $\sigma_1^2 \neq \sigma_2^2$.

$$H_0: \sigma_1^2 = \sigma_2^2 \quad H_a: \sigma_1^2 \neq \sigma_2^2$$

$$F_c = \frac{s_1^2}{s_2^2} = \frac{23.55}{40.24} = 0.5852$$

$$df_1 = n_1 - 1 = 7 \quad df_2 = n_2 - 1 = 11$$

Rejection region: $F > F_{0.025,7,11}$ or $F < F_{0.975,7,11}$

$$F_{0.025,7,11} = 3.76 \quad F_{0.975,7,11} = \frac{1}{F_{0.025,11,7}} = \frac{1}{4.7} = 0.21$$

Since F_c is less than 3.76 and greater than 0.21 we fail to reject H_0 at the $\alpha = 0.05$ level, and conclude that the population variances are equal.

- (b) (10) State the appropriate hypotheses to determine if the scores indicate that the new method is an improvement over the standard method. Compute the test statistic and an approximate p-value and use it to make your decision.

$$H_0: \mu_1 - \mu_2 \leq 0 \quad \text{vs} \quad H_a: \mu_1 - \mu_2 > 0$$

$$s_p^2 = \frac{7(23.55) + 11(40.24)}{8+12-2} = 33.75 \quad s_p = \sqrt{s_p^2} = 5.81$$

$$t_c = \frac{(\bar{y}_1 - \bar{y}_2) - 0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{77.125 - 72.333}{5.81 \sqrt{\frac{1}{8} + \frac{1}{12}}} = \frac{4.792}{2.6516} = 1.81$$

Looking at the t-table for $df = n_1 + n_2 - 2 = 8 + 12 - 2 = 18$, we have $0.025 < p < 0.05 \Rightarrow$ Reject H_0 at $\alpha = 0.05$

Since $p < 0.05$. Conclude that the new method is an improvement over the standard method.

- (c) (10) Construct a 95% confidence interval for $\mu_1 - \mu_2$. Show work. Explain what the experimenter needs to have done to make this estimate more accurate (i.e., to obtain a narrower interval)? Use the standard error of the difference in means to explain.

95% CI for $\mu_1 - \mu_2$:

$$(\bar{y}_1 - \bar{y}_2) \pm t_{0.025, 18} \cdot s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$4.79 \pm (2.101)(5.81) \sqrt{\frac{1}{8} + \frac{1}{12}}$$

2.6516

$$4.79 \pm 5.57$$

$$CI: (-0.78, 10.36)$$

As the sample size increases $t_{0.025, df}$ decreases and $se(\bar{y}_1 - \bar{y}_2) = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$ decreases $\Rightarrow$ CI will become narrower and more accurate.

2. A consumer protection magazine was interested in comparing tires purchased from four different companies that each claimed their tires would last 40,000 miles. A random sample of 10 tires of each brand was obtained and tested under simulated road conditions. The number of miles until the tread thickness reached a specified depth was recorded for all tires. The data are given (in 1,000 miles) below:

Brand	Miles(in thousands)										$\sum y$	$\bar{y}$
I	38.9	39.7	42.3	39.5	39.6	35.6	36.0	39.2	37.6	39.5	387.9	38.79
II	44.6	46.9	48.7	41.5	37.5	33.1	43.4	36.5	32.5	42.0	406.7	40.67
III	51.3	53.6	58.2	52.1	51.2	53.1	51.8	55.8	52.1	51.4	530.6	53.06
IV	39.1	38.7	40.4	41.9	39.4	36.1	40.5	35.2	32.3	36.0	379.6	37.96

Part of the ANOVA table is shown below:

Source	df	SS	MS	F	p-value
Treatment	$t-1 = 3$	1491.81	$SS/df = 497.27$	$MS/MSE = 40.66$	1.188×10^{-11}
Error	$n-t = 36$	440.28	$SSE/df = 12.23$	—	—
Total	$n-1 = 39$	1932.09	—	—	—

- (a) (10) Fill in the missing values in the ANOVA table (include both equations and actual number).
- (b) (10) Write out the linear statistical model and describe it, including **all** assumptions. Be specific!

$y_{ij} = \mu + \alpha_i + \epsilon_{ij}$ where y_{ij} is the j^{th} observation in the i^{th} group, μ is the overall mean, α_i is the brand effect of the i^{th} brand/group and ϵ_{ij} is the random error term for individual j in the i^{th} group.

Assumptions: $\epsilon_{ij} \sim N(0, \sigma^2)$
 $\uparrow \uparrow \nwarrow$
 normal independent homogeneous variance

- (c) (3) What type of experimental design is it? How do you know? Justify your answer.

Completely randomized design (CRD).

We have 10 random tires in 4 groups.

- (d) (2) What hypothesis can you test with the F and p-values found in the ANOVA table? Write down the null and alternative hypotheses.

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$$

H_a : at least one of the μ 's is not equal to the others

- (e) (5) Using the p-value given in the ANOVA table, state your decision and conclusion in terms of the problem.

$$p\text{-value} = 1.188 \cdot 10^{-11} \quad \alpha = 0.05$$

Since $p\text{-value} < \alpha$ we reject H_0 and conclude that at least one of the population means is different from the others.

- (f) (3) The company producing *Brand I* is located in Nebraska. We are interested in comparing the Nebraska tires with all of the other tires. Write a single linear contrast that describes the comparison described above.

$$l = \mu_1 - \frac{\mu_2 + \mu_3 + \mu_4}{3}$$

- (g) (3) Show that the linear combination you specified in part (f) is a linear contrast.

$$a_1 = 3 \quad a_2 = -1 \quad a_3 = -1 \quad a_4 = -1$$

$$\sum_{i=1}^4 a_i = 3 - 1 - 1 - 1 = 0 \quad \checkmark$$

(h) (10) Using the t-distribution, test whether the linear contrast described in part (f) is significant. Make sure you include every step of the test. $\alpha = 0.05$

$$H_0: \ell = 0 \quad H_a: \ell \neq 0$$

$$t_c = \frac{\hat{\ell}}{\sqrt{V(\hat{\ell})}}$$

$$t_c = \frac{60.33}{\sqrt{14.676}} = 15.748$$

$$\hat{\ell} = 3(38.79) - 1(40.67) - 1(53.06) - 1(37.96) = 60.33$$

$$V(\hat{\ell}) = s_w^2 \cdot \sum_{i=1}^4 \frac{a_i^2}{n_i} = 12.23 \cdot \frac{(9+1+1+1)}{10}$$

$$= 12.23 \cdot 1.2 = 14.676$$

$$t_{\alpha/2, df_{error}} = 0.025, 36 = 2.03 \quad (\text{using } df=35 \text{ because table doesn't have } t \text{ value for } df=36)$$

Since $t_c > t_{\alpha/2, df}$ we reject H_0 and conclude that there is a difference between the NE brand and the average of the other brands.

(i) (10) Perform Tukey's procedure to perform every pairwise comparison.

$$\alpha = 0.05$$

	1	2	3	4
	Brand IV	Brand I	Brand II	Brand III
$\bar{y}_i$	37.96	38.79	40.67	53.06

$$W = q_{\alpha}(t, r) \sqrt{\frac{s_w^2}{n}}$$

$$W = 3.79 \cdot \sqrt{\frac{12.23}{10}}$$

$$W = 4.19$$

$$t = 4 \quad r = 36 \Rightarrow q_{0.05}(4, 36) = 3.79$$

(using $df=40$ from table)

$$s_w^2 = \text{MSE} = 12.23$$

$$n = 10$$

1	2	3	4



We can simplify:

1	2	3	4

$$53.06 - 37.96 = 15.1 > W$$

$$40.67 - 37.96 = 2.71 < W$$

$$53.06 - 38.79 = 14.27 > W$$

$$40.67 - 38.79 = 1.88 < W$$

$$53.06 - 40.67 = 12.39 > W$$

Conclusion:

Brand III is significantly different from brand I, brand II and brand IV, but brands I, II, and IV are not significantly different from each other.

3. TRUE/FALSE. Circle your answer. (2 points each.)

- (a) T / ☒ F Sampling error is the only cause two sample means are different.
- (b) T / ☒ F The number of mutually orthogonal contrasts is the same as the number of means.
- (c) ☒ T / F A significant F-test from a one-way ANOVA tells us that there is a significant difference somewhere among the means, but it doesn't tell us where it is.
- (d) T / ☒ F Within-group variance is caused by sampling error and treatment effects.
- (e) ☒ T / F The F statistic in the ANOVA table compares the ratio of between-group variance to within-group variance.
- (f) T / ☒ F The paired t-test assumes that the data in each group are normally distributed.
- (g) T / ☒ F The paired t-test is appropriate for comparing the means of independent groups of observations.